

prompt

September 23, 2026

0.0.1 Required Assignment 5.1: Will the Customer Accept the Coupon?

Context

Imagine driving through town and a coupon is delivered to your cell phone for a restaurant near where you are driving. Would you accept that coupon and take a short detour to the restaurant? Would you accept the coupon but use it on a subsequent trip? Would you ignore the coupon entirely? What if the coupon was for a bar instead of a restaurant? What about a coffee house? Would you accept a bar coupon with a minor passenger in the car? What about if it was just you and your partner in the car? Would weather impact the rate of acceptance? What about the time of day?

Obviously, proximity to the business is a factor on whether the coupon is delivered to the driver or not, but what are the factors that determine whether a driver accepts the coupon once it is delivered to them? How would you determine whether a driver is likely to accept a coupon?

Overview

The goal of this project is to use what you know about visualizations and probability distributions to distinguish between customers who accepted a driving coupon versus those that did not.

Data

This data comes to us from the UCI Machine Learning repository and was collected via a survey on Amazon Mechanical Turk. The survey describes different driving scenarios including the destination, current time, weather, passenger, etc., and then ask the person whether he will accept the coupon if he is the driver. Answers that the user will drive there 'right away' or 'later before the coupon expires' are labeled as ' $Y = 1$ ' and answers 'no, I do not want the coupon' are labeled as ' $Y = 0$ '. There are five different types of coupons – less expensive restaurants (under \$20), coffee houses, carry out & take away, bar, and more expensive restaurants (\$20 - \$50).

Deliverables

Your final product should be a brief report that highlights the differences between customers who did and did not accept the coupons. To explore the data you will utilize your knowledge of plotting, statistical summaries, and visualization using Python. You will publish your findings in a public facing github repository as your first portfolio piece.

0.0.2 Data Description

Keep in mind that these values mentioned below are average values.

The attributes of this data set include: 1. User attributes - Gender: male, female - Age: below 21, 21 to 25, 26 to 30, etc. - Marital Status: single, married partner, unmarried partner, or widowed - Number of children: 0, 1, or more than 1 - Education: high school, bachelors degree, associates degree, or graduate degree - Occupation: architecture & engineering, business & financial, etc. - Annual income: less than \\$12500, \\$12500 - \\$24999, \\$25000 - \\$37499, etc. - Number of times that he/she goes to a bar: 0, less than 1, 1 to 3, 4 to 8 or greater than 8 - Number of times that he/she buys takeaway food: 0, less than 1, 1 to 3, 4 to 8 or greater than 8 - Number of times that he/she goes to a coffee house: 0, less than 1, 1 to 3, 4 to 8 or greater than 8 - Number of times that he/she eats at a restaurant with average expense less than \\$20 per person: 0, less than 1, 1 to 3, 4 to 8 or greater than 8 - Number of times that he/she goes to a bar: 0, less than 1, 1 to 3, 4 to 8 or greater than 8

2. Contextual attributes

- Driving destination: home, work, or no urgent destination
- Location of user, coupon and destination: we provide a map to show the geographical location of the user, destination, and the venue, and we mark the distance between each two places with time of driving. The user can see whether the venue is in the same direction as the destination.
- Weather: sunny, rainy, or snowy
- Temperature: 30F, 55F, or 80F
- Time: 10AM, 2PM, or 6PM
- Passenger: alone, partner, kid(s), or friend(s)

3. Coupon attributes

- time before it expires: 2 hours or one day

```
[7]: import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd
import numpy as np
```

0.0.3 Problems

Use the prompts below to get started with your data analysis.

1. Read in the `coupons.csv` file.

```
[18]: data = pd.read_csv('data/coupons.csv')
```

```
[19]: data.head()
```

```
[19]:
```

	destination	passanger	weather	temperature	time	\
0	No Urgent Place	Alone	Sunny	55	2PM	
1	No Urgent Place	Friend(s)	Sunny	80	10AM	
2	No Urgent Place	Friend(s)	Sunny	80	10AM	
3	No Urgent Place	Friend(s)	Sunny	80	2PM	
4	No Urgent Place	Friend(s)	Sunny	80	2PM	

	coupon expiration	gender	age	maritalStatus	...	\
0	Restaurant(<20)	1d	Female	21	Unmarried partner	...

1	Coffee House	2h	Female	21	Unmarried partner	...
2	Carry out & Take away	2h	Female	21	Unmarried partner	...
3	Coffee House	2h	Female	21	Unmarried partner	...
4	Coffee House	1d	Female	21	Unmarried partner	...

	CoffeeHouse	CarryAway	RestaurantLessThan20	Restaurant20To50	\
0	never	NaN	4~8	1~3	
1	never	NaN	4~8	1~3	
2	never	NaN	4~8	1~3	
3	never	NaN	4~8	1~3	
4	never	NaN	4~8	1~3	

	toCoupon_GEQ5min	toCoupon_GEQ15min	toCoupon_GEQ25min	direction_same	\
0	1	0	0	0	
1	1	0	0	0	
2	1	1	0	0	
3	1	1	0	0	
4	1	1	0	0	

	direction_opp	Y
0	1	1
1	1	0
2	1	1
3	1	0
4	1	0

[5 rows x 26 columns]

2. Investigate the dataset for missing or problematic data.

```
[12]: #data.describe()
categorical_cols = data.select_dtypes(include=['category', 'object']).columns.
      ↪ tolist()
data[categorical_cols] = data[categorical_cols].astype('category')
print("\n data types ---")
data.info()
print("\n--- Missing Values ---")
data1 = data.isnull().sum()[data.isnull().sum() > 0]
print(data1)
print("\n--- Missing Values % ---")
print((data.isnull().sum() / len(data) * 100).sort_values(ascending=False))

#print(data.dtypes)

print("\n--- Categorical Values ---")
for col in data.select_dtypes(include="category").columns:
    print(f"\n{col}:")
```

```

print(data[col].value_counts(dropna=False))

# Duplicate rows
print("\n Duplicate Rows -")
print(data.duplicated(keep='first').sum())
data2 = data[data.duplicated()]

```

```

data types ---
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 12684 entries, 0 to 12683
Data columns (total 26 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   destination                          12684 non-null  category
1   passanger                           12684 non-null  category
2   weather                             12684 non-null  category
3   temperature                         12684 non-null  int64
4   time                                12684 non-null  category
5   coupon                             12684 non-null  category
6   expiration                          12684 non-null  category
7   gender                              12684 non-null  category
8   age                                 12684 non-null  category
9   maritalStatus                      12684 non-null  category
10  has_children                        12684 non-null  int64
11  education                          12684 non-null  category
12  occupation                         12684 non-null  category
13  income                             12684 non-null  category
14  car                                 108 non-null    category
15  Bar                                 12577 non-null  category
16  CoffeeHouse                        12467 non-null  category
17  CarryAway                          12533 non-null  category
18  RestaurantLessThan20               12554 non-null  category
19  Restaurant20To50                   12495 non-null  category
20  toCoupon_GEQ5min                   12684 non-null  int64
21  toCoupon_GEQ15min                  12684 non-null  int64
22  toCoupon_GEQ25min                  12684 non-null  int64
23  direction_same                     12684 non-null  int64
24  direction_opp                      12684 non-null  int64
25  Y                                   12684 non-null  int64
dtypes: category(18), int64(8)
memory usage: 1020.1 KB

--- Missing Values ---
car                12576
Bar                 107
CoffeeHouse        217

```

```
CarryAway          151
RestaurantLessThan20 130
Restaurant20To50    189
dtype: int64
```

--- Missing Values % ---

```
car          99.148534
CoffeeHouse   1.710817
Restaurant20To50 1.490066
CarryAway     1.190476
RestaurantLessThan20 1.024913
Bar           0.843582
destination   0.000000
passanger     0.000000
direction_opp 0.000000
direction_same 0.000000
toCoupon_GEQ25min 0.000000
toCoupon_GEQ15min 0.000000
toCoupon_GEQ5min 0.000000
income        0.000000
occupation    0.000000
education     0.000000
has_children  0.000000
maritalStatus 0.000000
age           0.000000
gender        0.000000
expiration    0.000000
coupon        0.000000
time          0.000000
temperature   0.000000
weather       0.000000
Y             0.000000
dtype: float64
```

--- Categorical Values ---

destination:

```
No Urgent Place  6283
Home             3237
Work             3164
```

Name: destination, dtype: int64

passanger:

```
Alone      7305
Friend(s)  3298
Partner    1075
Kid(s)     1006
```

Name: passanger, dtype: int64

weather:
Sunny 10069
Snowy 1405
Rainy 1210
Name: weather, dtype: int64

time:
6PM 3230
7AM 3164
10AM 2275
2PM 2009
10PM 2006
Name: time, dtype: int64

coupon:
Coffee House 3996
Restaurant(<20) 2786
Carry out & Take away 2393
Bar 2017
Restaurant(20-50) 1492
Name: coupon, dtype: int64

expiration:
1d 7091
2h 5593
Name: expiration, dtype: int64

gender:
Female 6511
Male 6173
Name: gender, dtype: int64

age:
21 2653
26 2559
31 2039
50plus 1788
36 1319
41 1093
46 686
below21 547
Name: age, dtype: int64

maritalStatus:
Married partner 5100
Single 4752
Unmarried partner 2186

Divorced	516
Widowed	130

Name: maritalStatus, dtype: int64

education:

Some college - no degree	4351
Bachelors degree	4335
Graduate degree (Masters or Doctorate)	1852
Associates degree	1153
High School Graduate	905
Some High School	88

Name: education, dtype: int64

occupation:

Unemployed	1870
Student	1584
Computer & Mathematical	1408
Sales & Related	1093
Education&Training&Library	943
Management	838
Office & Administrative Support	639
Arts Design Entertainment Sports & Media	629
Business & Financial	544
Retired	495
Food Preparation & Serving Related	298
Healthcare Practitioners & Technical	244
Healthcare Support	242
Community & Social Services	241
Legal	219
Transportation & Material Moving	218
Protective Service	175
Architecture & Engineering	175
Personal Care & Service	175
Life Physical Social Science	170
Construction & Extraction	154
Installation Maintenance & Repair	133
Production Occupations	110
Building & Grounds Cleaning & Maintenance	44
Farming Fishing & Forestry	43

Name: occupation, dtype: int64

income:

\$25000 - \$37499	2013
\$12500 - \$24999	1831
\$37500 - \$49999	1805
\$100000 or More	1736
\$50000 - \$62499	1659
Less than \$12500	1042

\$87500 - \$99999	895
\$75000 - \$87499	857
\$62500 - \$74999	846

Name: income, dtype: int64

car:	
NaN	12576
Mazda5	22
Scooter and motorcycle	22
do not drive	22
Car that is too old to install Onstar :D	21
crossover	21

Name: car, dtype: int64

Bar:	
never	5197
less1	3482
1~3	2473
4~8	1076
gt8	349
NaN	107

Name: Bar, dtype: int64

CoffeeHouse:	
less1	3385
1~3	3225
never	2962
4~8	1784
gt8	1111
NaN	217

Name: CoffeeHouse, dtype: int64

CarryAway:	
1~3	4672
4~8	4258
less1	1856
gt8	1594
never	153
NaN	151

Name: CarryAway, dtype: int64

RestaurantLessThan20:	
1~3	5376
4~8	3580
less1	2093
gt8	1285
never	220
NaN	130

Name: RestaurantLessThan20, dtype: int64

Restaurant20To50:

less1	6077
1~3	3290
never	2136
4~8	728
gt8	264
NaN	189

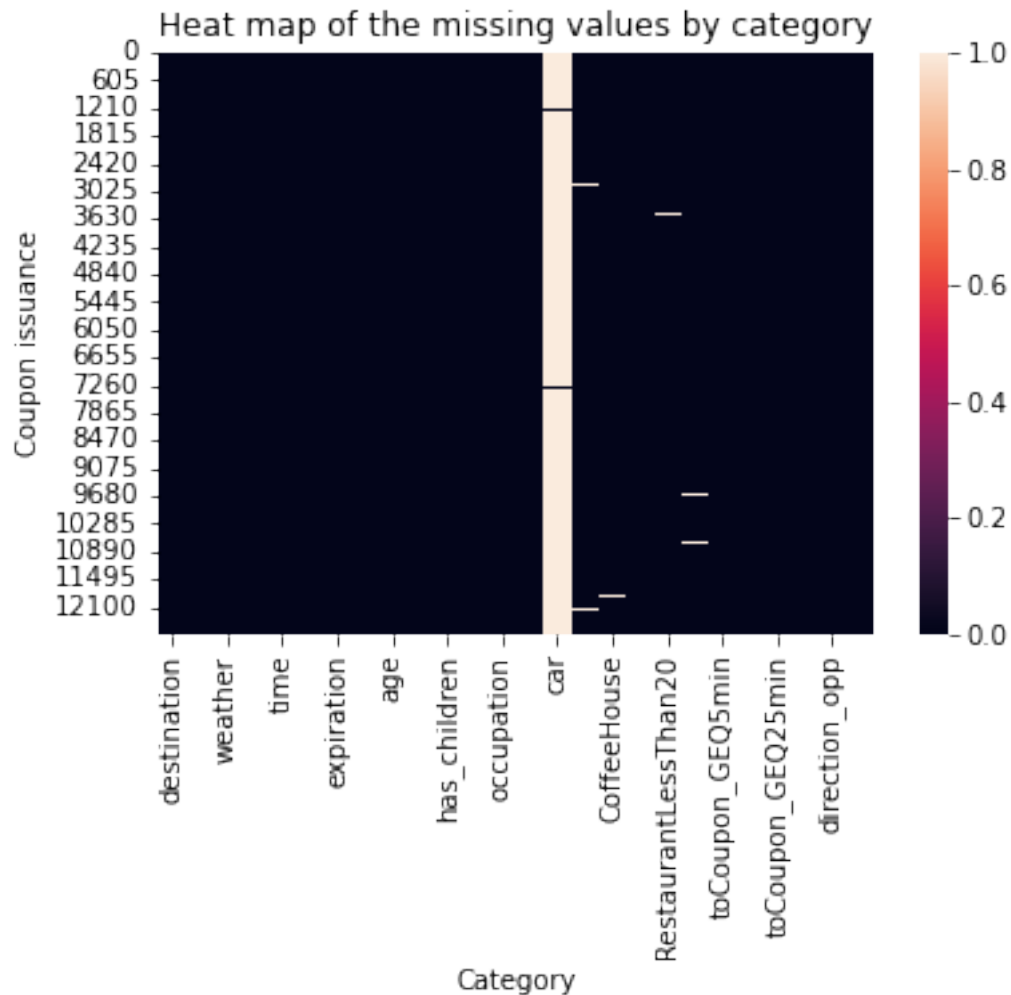
Name: Restaurant20To50, dtype: int64

Duplicate Rows -
74

3. Decide what to do about your missing data – drop, replace, other...

```
[47]: sns.heatmap(data.isna());  
plt.title('Heat map of the missing values by category')  
plt.xlabel('Category')  
plt.ylabel('Coupon issuance')  
  
# The column "Car" can be dropped as 99% of the data is missing in the missing_  
# values% above.  
  
# The categorical Objects 'Bar', 'CoffeeHouse', 'Restaurant20To50',  
# 'CarryAway', 'RestaurantLessThan20' contribute to  
# approx 1% of the total data set each. A majority of these should be left null_  
# or filled with Nan since these  
# missing values correspond to those categories that are given a different_  
# category coupon than itself.  
# For e.g. Bar object is blank or missing when the coupon given is CoffeeHouse.  
  
# only the small subset of each of the above mentioned 5 categories that have_  
# missing values when given the  
# respective coupon itself should be replaced with their most used category so_  
# that there is no loss in data  
# or a major bias.
```

```
[47]: Text(33.0, 0.5, 'Coupon issuance')
```



4. What proportion of the total observations chose to accept the coupon?

```
[33]: total = data['Y'].count()
      #Drivers who accepted the coupon per category
      acceptedTotal = data.loc[data['Y'] == 1].count()
      print(acceptedTotal/total)

      acceptance_rate = data['Y'].mean() * 100
      print('\n acceptance rate is')
      print(acceptance_rate )
```

```
destination      0.568433
passanger        0.568433
weather          0.568433
temperature      0.568433
time             0.568433
```

coupon	0.568433
expiration	0.568433
gender	0.568433
age	0.568433
maritalStatus	0.568433
has_children	0.568433
education	0.568433
occupation	0.568433
income	0.568433
car	0.004888
Bar	0.564096
CoffeeHouse	0.559760
CarryAway	0.560470
RestaurantLessThan20	0.561416
Restaurant20To50	0.560155
toCoupon_GEQ5min	0.568433
toCoupon_GEQ15min	0.568433
toCoupon_GEQ25min	0.568433
direction_same	0.568433
direction_opp	0.568433
Y	0.568433
dtype: float64	

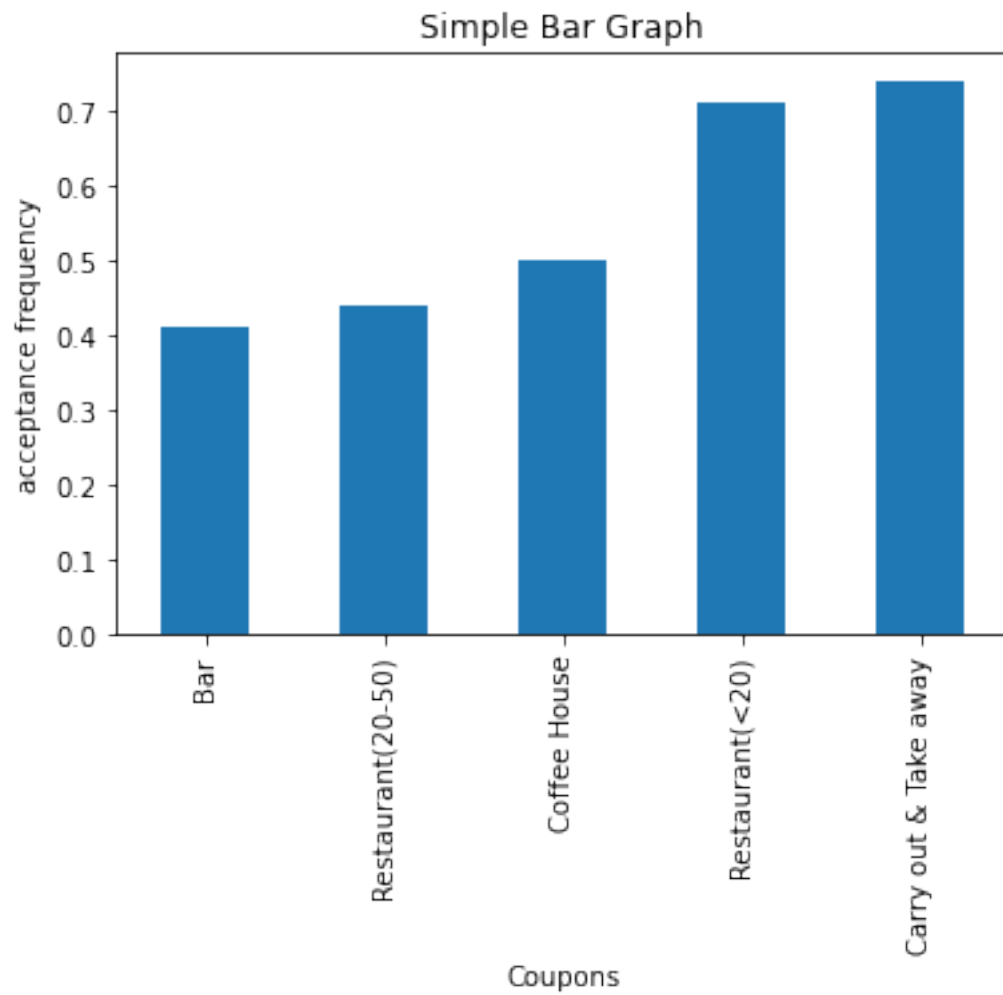
acceptance rate is
56.84326710816777

[]:

5. Use a bar plot to visualize the `coupon` column.

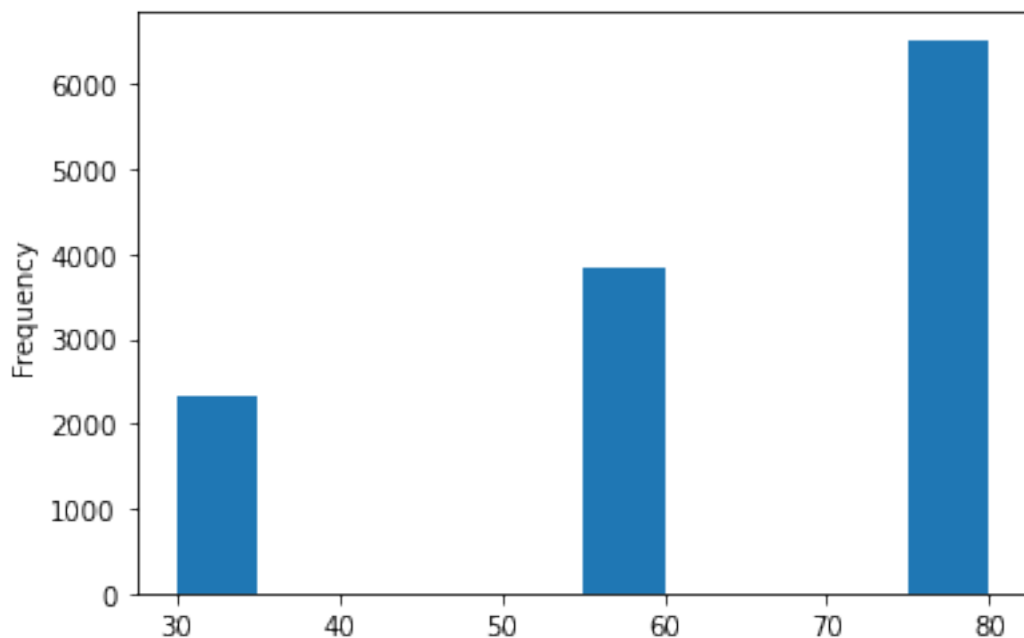
```
[14]: #plt.bar("coupons", "Y")
data.groupby("coupon")["Y"].mean().round(2).sort_values().plot(kind = "bar")
plt.title('Simple Bar Graph')
plt.xlabel('Coupons')
plt.ylabel('acceptance frequency')
```

```
[14]: Text(0, 0.5, 'acceptance frequency')
```



6. Use a histogram to visualize the temperature column.

```
[15]: data['temperature'].plot(kind = 'hist');
```



Investigating the Bar Coupons

Now, we will lead you through an exploration of just the bar related coupons.

1. Create a new DataFrame that contains just the bar coupons.

```
[37]: #df is the data set with only Bar coupons
df = data[data['coupon'] == 'Bar']
df.head()
```

```
[37]:      destination  passanger weather  temperature  time coupon expiration \
9    No Urgent Place    Kid(s)   Sunny           80  10AM    Bar          1d
13              Home      Alone   Sunny           55   6PM    Bar          1d
17              Work      Alone   Sunny           55   7AM    Bar          1d
24  No Urgent Place  Friend(s)   Sunny           80  10AM    Bar          1d
35              Home      Alone   Sunny           55   6PM    Bar          1d
```

```
      gender age  maritalStatus  ...  CoffeeHouse  CarryAway  \
9   Female  21  Unmarried partner  ...      never      NaN
13  Female  21  Unmarried partner  ...      never      NaN
17  Female  21  Unmarried partner  ...      never      NaN
24   Male  21           Single  ...     less1     4~8
35   Male  21           Single  ...     less1     4~8
```

```
      RestaurantLessThan20  Restaurant20To50  toCoupon_GEQ5min  toCoupon_GEQ15min  \
9              4~8              1~3              1              1
13             4~8              1~3              1              0
```

17	4~8	1~3	1	1
24	4~8	less1	1	0
35	4~8	less1	1	0

	toCoupon_GEQ25min	direction_same	direction_opp	Y
9	0	0	1	0
13	0	1	0	1
17	1	0	1	0
24	0	0	1	1
35	0	1	0	1

[5 rows x 26 columns]

2. What proportion of bar coupons were accepted?

```
[45]: acceptance_rate = df['Y'].mean() * 100
print('\n acceptance_rate' , acceptance_rate)

bar_total = df['Y'].count()
print('\n total bar coupons received', bar_total)
#Drivers who accepted the coupon
acceptedTotal_bar = df.loc[df['Y'] == 1].count()
print('\n')
print(acceptedTotal_bar/bar_total)
```

acceptance_rate 41.00148735746158

total bar coupons received 2017

destination	0.410015
passanger	0.410015
weather	0.410015
temperature	0.410015
time	0.410015
coupon	0.410015
expiration	0.410015
gender	0.410015
age	0.410015
maritalStatus	0.410015
has_children	0.410015
education	0.410015
occupation	0.410015
income	0.410015
car	0.001487
Bar	0.406049
CoffeeHouse	0.403570

```
CarryAway          0.402082
RestaurantLessThan20 0.404065
Restaurant20To50    0.404065
toCoupon_GEQ5min    0.410015
toCoupon_GEQ15min   0.410015
toCoupon_GEQ25min   0.410015
direction_same      0.410015
direction_opp       0.410015
Y                   0.410015
dtype: float64
```

3. Compare the acceptance rate between those who went to a bar 3 or fewer times a month to those who went more.

```
[44]: less = df[df['Bar'].isin(['never', 'less1', '1~3'])]
morethan3 = df[df['Bar'].isin(['4~8', 'gt8'])]
acceptance_rate_3_or_less = less['Y'].mean() * 100
acceptance_rate_more_than_3 = morethan3['Y'].mean() * 100
print("\n less than 3:")
print(acceptance_rate_3_or_less)
print("\n more than 3:")
print(acceptance_rate_more_than_3)

# The coupon acceptance rate for those who went more than 3 times to a bar is
↳ ~39% absolute difference higher than those who went 3 or fewer times a month.
↳
```

```
less than 3:
37.061769616026716
```

```
more than 3:
76.88442211055276
```

4. Compare the acceptance rate between drivers who go to a bar more than once a month and are over the age of 25 to the all others. Is there a difference?

```
[116]: bar_more_than_once_over25 = df[
        (df['Bar'].isin(['1~3', '4~8', 'gt8'])) &
        (df['age'].isin(['26-30', '31-36', '41-50', '50plus']))]

bar_all_others = df.drop(bar_more_than_once_over25.index)

print("More than once/month & over 25:")
acc_rate = bar_more_than_once_over25['Y'].mean() * 100
print(round(acc_rate,2))
print("All others:")
acc_rate_others = bar_all_others['Y'].mean() * 100
```

```
print(round(acc_rate_others,2))

# From the solution obtained we see that those who went more than once a month
↳ to a bar
# and are over an age of 25 years are 27% absolute difference higher than the
↳ rest.
```

More than once/month & over 25:

67.16

All others:

40.1

5. Use the same process to compare the acceptance rate between drivers who go to bars more than once a month and had passengers that were not a kid and had occupations other than farming, fishing, or forestry.

```
[120]: New_df = df[
    (df['Bar'].isin(['1~3', '4~8', 'gt8'])) &
    (df['passanger'] != 'Kid(s))' &
    (df['occupation'] != 'Farming Fishing & Forestry')]

New_all_others = df.drop(New_df.index)

acc_rate_Newdf = New_df['Y'].mean() * 100
print("drivers who go to bars more than once a month and had passengers that
↳ were not a kid and had occupations")
print(round(acc_rate_Newdf,2))
acc_rate_Newothers = New_all_others['Y'].mean() * 100
print("all other drivers")
print(round(acc_rate_Newothers,2))

# From the solution obtained we see that those who went more than once a month
↳ to a bar and had passengers that were not a kid
# and had occupations other than farming, fishing, or forestry are likely to
↳ accept the coupons by 41% absolute difference higher than the rest.
```

drivers who go to bars more than once a month and had passengers that were not a kid and had occupations

71.32

all other drivers

29.6

6. Compare the acceptance rates between those drivers who:
 - go to bars more than once a month, had passengers that were not a kid, and were not widowed
OR
 - go to bars more than once a month and are under the age of 30 OR
 - go to cheap restaurants more than 4 times a month and income is less than 50K.

```

[46]: # Acceptance rate of drivers who go more than once a month, had passengers that
      ↳ were not a kid, and were not widowed
df2 = df[
    (df['Bar'].isin(['1~3', '4~8', 'gt8'])) &
    (df['passanger'] != 'Kid(s))' &
    (df['occupation'] != 'Farming Fishing & Forestry')]

df2_all_others = df.drop(df2.index)

acc_rate_df2 = df2['Y'].mean() * 100
print("acc rate of those who go to bars more than once a month, had passengers
      ↳ that were not a kid, and were not widowed ")
print(round(acc_rate_df2,2))
acc_rate_df2others = df2_all_others['Y'].mean() * 100

# Acceptance rate of drivers who go more than once a month and are Under 30
df3 = df[
    (df['Bar'].isin(['1~3', '4~8', 'gt8'])) &
    (df['age'].isin(['below21', '21']))]
df3_acceptance_rate = df3['Y'].mean() * 100
print("acc rate of those who go to bars more than once a month and are under
      ↳ the age of 30")
print(round(df3_acceptance_rate,2))

# Acceptance rate of drivers who go to cheap restaurants more than 4 times a
      ↳ month and income is less than 50K
df4 = df[
    (df['RestaurantLessThan20'].isin(['4~8', 'gt8'])) &
    (df['income'].isin([
        'Less than $12500',
        '$12500 - $24999',
        '$25000 - $37499',
        '$37500 - $49999']))]
df4_acceptance_rate = df4['Y'].mean() * 100
print("acceptance rate of those who go to cheap restaurants more than 4 times a
      ↳ month and income is less than 50K")
print(round(df4_acceptance_rate,2))

# Acceptance rate of drivers who go more than once a month, had passengers that
      ↳ were not a kid, and were not widowed
# is 4% absolute difference higher than those who are under the age of 30; and
      ↳ are 26% absolute difference higher than those
# who go to cheap restaurants more than 4 times a month and income is less than
      ↳ 50 K.

```

acc rate of those who go to bars more than once a month, had passengers that
were not a kid, and were not widowed

71.32

acc rate of those who go to bars more than once a month and are under the age of 30

67.05

acceptance rate of those who go to cheap restaurants more than 4 times a month and income is less than 50K

45.35

7. Based on these observations, what do you hypothesize about drivers who accepted the bar coupons?

```
[ ]: Based on these observations and associations , further statistical analysis
    ↳would be needed.

#-----Accepted Coupon-----:
#Drivers who frequently visit bars are more likely to accept bar coupons than
    ↳drivers who rarely or never visit bars.
#Drivers who accepted bar coupons are more likely to be frequent bar visitors,
    ↳and their acceptance behavior may also
# be associated with age, passenger characteristics, and occupation and
    ↳existing location.
#Bar coupons are least accepted coupons among others that were offered.

#-----Not Accepted coupon-----:
#Drivers who did not accept coupons tend to be associated with less frequent
    ↳bar visits, driving alone or with children,
#older age groups, short coupon expiration periods, unfavorable weather
    ↳conditions, and certain times of day.
# Non-acceptance is relatively high Drivers for Bar coupons, with approximately
    ↳59% compared to other coupons.
```

0.0.4 Independent Investigation

Using the bar coupon example as motivation, you are to explore one of the other coupon groups and try to determine the characteristics of passengers who accept the coupons.

[]:

[]:

[]:

[]:

[]: